

Dataset

1. Node Features (X_t and $x_{i,t}$)

These provide the real-time status of traffic at specific junctions or sensor locations.

- **Traffic Flow/Volume:** Number of vehicles passing the node per time interval.
- **Average Speed:** Mean velocity of vehicles at the node at time t .
- **Lane Occupancy:** Percentage of time the sensor is occupied, indicating density.
- **Queue Length:** Number of vehicles or length of the backup at the intersection.
- **Intersection Type:** Functional characteristics (e.g., signalized, roundabout, or priority-based).

2. Edge Features (E_t and $e_{ji,t}$)

These define the road segment properties connecting the nodes.

- **Road Capacity:** Theoretical maximum vehicle throughput for the segment.
- **Segment Length:** Physical distance between two nodes.
- **Road Quality Index:** Measured via the International Roughness Index (IRI) or Pavement Condition Index (PCI) to account for surface friction and damage.
- **Speed Limit:** Regulatory maximum speed for the segment.
- **Lane Count:** Number of available lanes for traffic flow.
- **Incident Flags:** Binary indicators for active accidents or road construction.

3. Weather Features (W_t and $w_{i,t}$)

Weather acts as a dynamic modulator for the node embeddings and the quantum update stage.

- **Precipitation Intensity:** Rainfall rates (mm/hr).
- **Visibility:** Distance of clear sight (meters), affected by fog or haze.
- **Temperature:** Ambient temperature ($^{\circ}\text{C}$), impacting road surface conditions.

4. Global Temporal Features (τ_t)

These capture the periodic and cyclic nature of urban mobility required for the $t \rightarrow t + 1$ transition.

- **Hour of Day:** Cyclic representation (Sine/Cosine transform) to preserve the 24-hour proximity.
- **Day of Week:** Indicator for weekday vs. weekend patterns.
- **Holiday Flag:** Binary indicator for public holidays or school breaks.
- **Special Events:** Flags for nearby major social gatherings (concerts, sporting events).

1. City as a Dynamic Graph

The foundation of the model represents the urban environment as a graph that evolves over discrete time intervals.

$$G_t = (V, E, X_t, W_t, \tau_t)$$

Explanation:

- G_t : Graph representation of the city at time t .
- V : Set of nodes (e.g., intersections, traffic signals).
- E : Set of edges (roads connecting intersections).
- X_t : Traffic feature matrix at time t (e.g., traffic density, queue length).
- W_t : Weather feature matrix at time t (e.g., precipitation, visibility).
- τ_t : Global temporal feature vector (e.g., hour of day, day of week).

This models the city as a dynamic system where traffic conditions are modulated by both infrastructure and environmental context.

2. Node Embedding (Contextualized)

Raw sensor data is transformed into a learned latent representation that incorporates external factors.

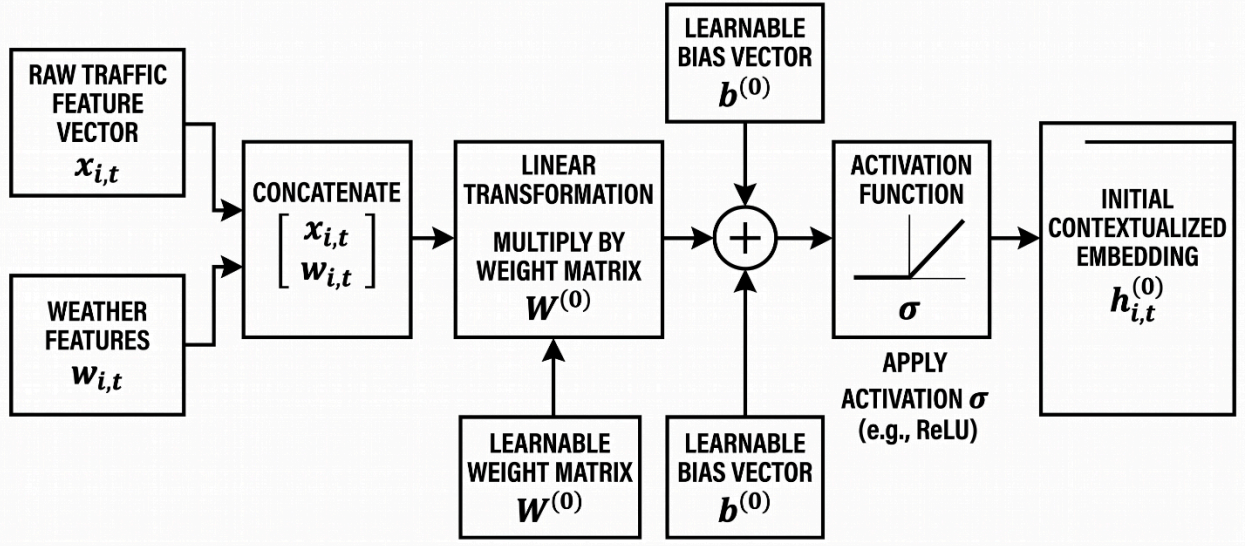
$$h_{i,t}^{(0)} = ReLU(W_v[x_{i,t} \parallel w_{i,t} \parallel \tau_t] + b_v)$$

Terms:

- $x_{i,t}$: Raw traffic feature vector of node i at time t .
- $w_{i,t}$: Weather features at node i at time t .
- W_v : Learnable weight matrix.
- b_v : Bias vector.
- $ReLU$: Activation function $\max(0, x)$.
- $h_{i,t}^{(0)}$: Initial contextualized embedding of node i at time t .

NODE EMBEDDING CALCULATION FLOW (EQUATION 2)

$$h_{i,t}^{(0)} = \sigma(W^{(0)} \cdot [x_{i,t}; w_{i,t}] + b^{(0)})$$



3. Message Passing (Spatio-Temporal Construction)

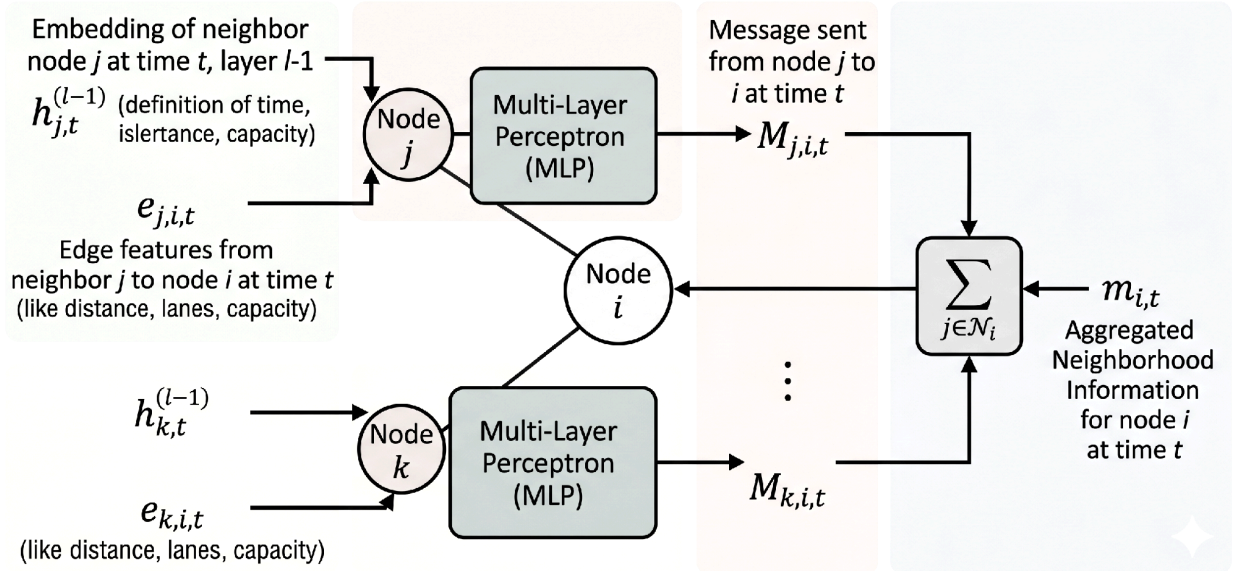
Each node receives information from its neighbors, weighted by the current road and environmental state.

$$m_{j \rightarrow i,t} = MLP(h_{j,t} \parallel e_{ji,t} \parallel w_{i,t} \parallel \tau_t)$$

Terms:

- $h_{j,t}$: Embedding of neighbor node j at time t .
- $e_{ji,t}$: Edge features from j to i (distance, lanes, current capacity) at time t .
- MLP : Multi-layer perceptron.
- $m_{j \rightarrow i,t}$: Message sent from node j to i at time t .

Message Passing (Spatio-Temporal Construction)



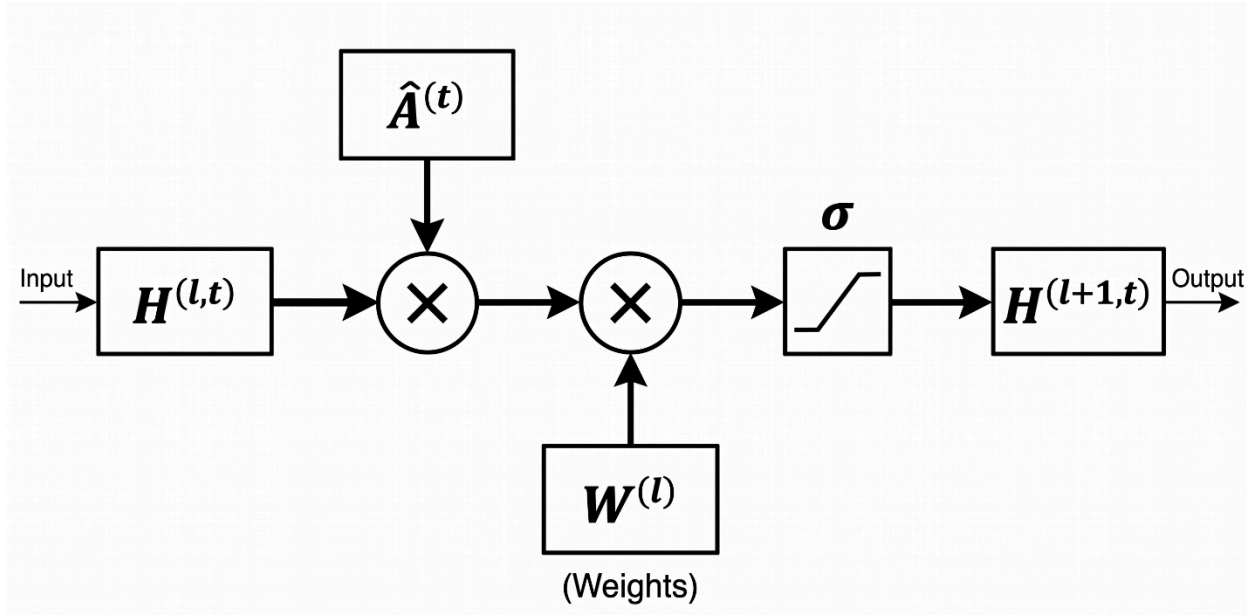
4. Graph Convolutional Update

The global state of the network is updated by propagating neighborhood information through the graph topology.

$$H_t^{(l+1)} = \sigma(A_t H_t^{(l)} W)$$

Explanation:

- $H_t^{(l)}$: Node features at layer l and time t .
- A_t : Adjacency matrix (connectivity) at time t .
- W : Learnable weights.
- σ : Activation function.



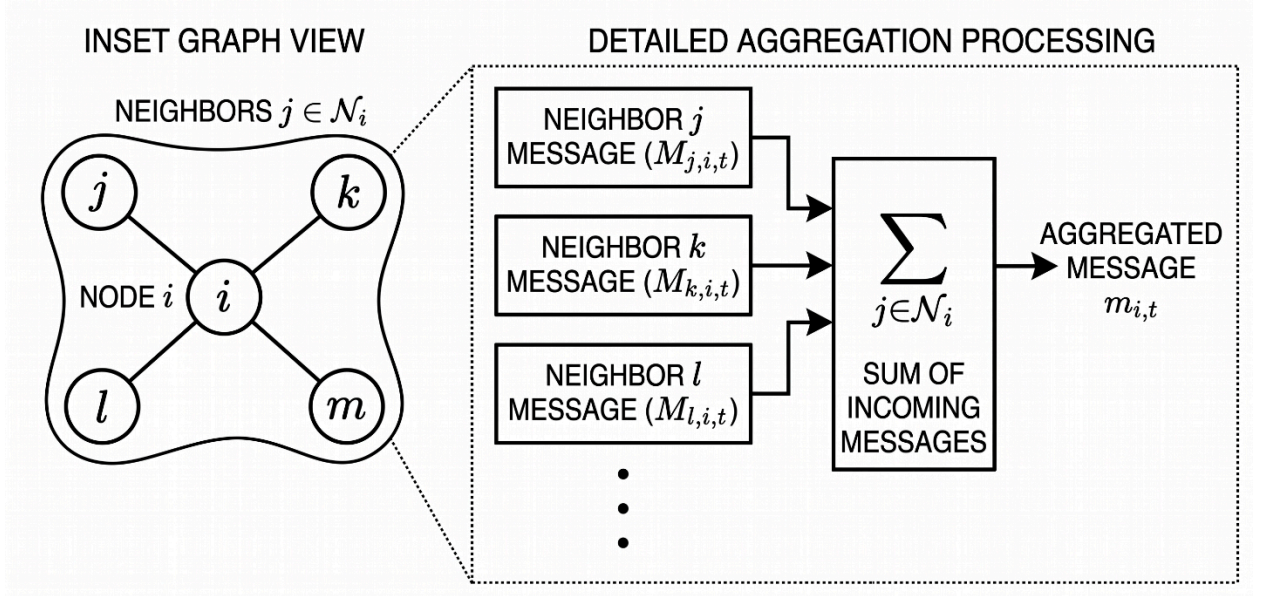
5. Neighborhood Aggregation

Node i sums the influences of all connected nodes to determine the local traffic "pressure."

$$a_{i,t} = \sum_{j \in N(i)} m_{j \rightarrow i,t}$$

Terms:

- $N(i)$: Set of neighbors of node i .
- $m_{j \rightarrow i,t}$: Messages from neighbors at time t .
- $a_{i,t}$: Aggregated message for node i at time t .



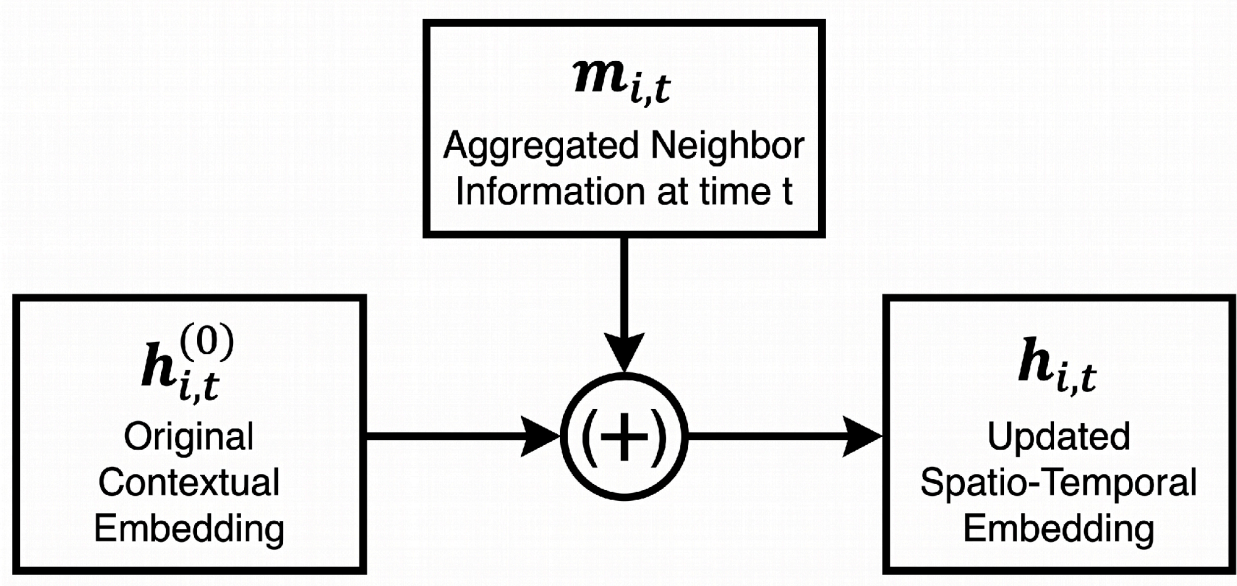
6. Residual Update

The model combines original node features with new neighborhood influence to prevent information loss.

$$h_{i,t}^{(1)} = h_{i,t}^{(0)} + a_{i,t}$$

Terms:

- $h_{i,t}^{(0)}$: Original contextual embedding.
- $a_{i,t}$: Aggregated neighbor information at time t .
- $h_{i,t}^{(1)}$: Updated spatio-temporal embedding.



7. Quantum Interface: Compression

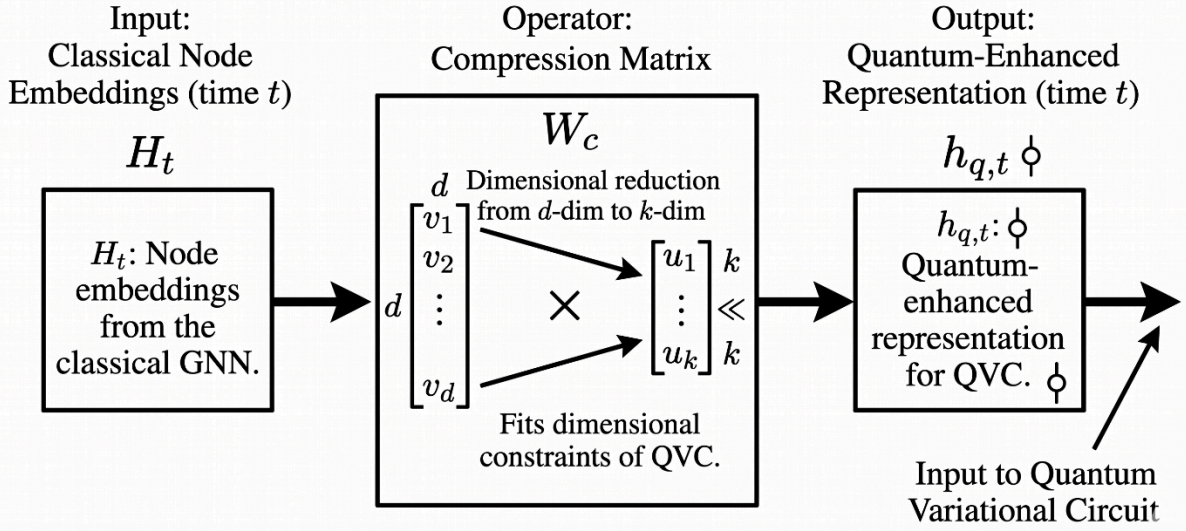
Classical features are compressed to fit the dimensional constraints of the quantum variational circuit.

$$Z_t = W_c H_t \rightarrow Z'_t = \mathcal{Q}_{\theta,t}(Z_t)$$

Explanation:

- H_t : Node embeddings from the classical GNN at time t .
- W_c : Compression matrix.
- Z'_t : Quantum-enhanced representation at time t .

Quantum Interface: Compression



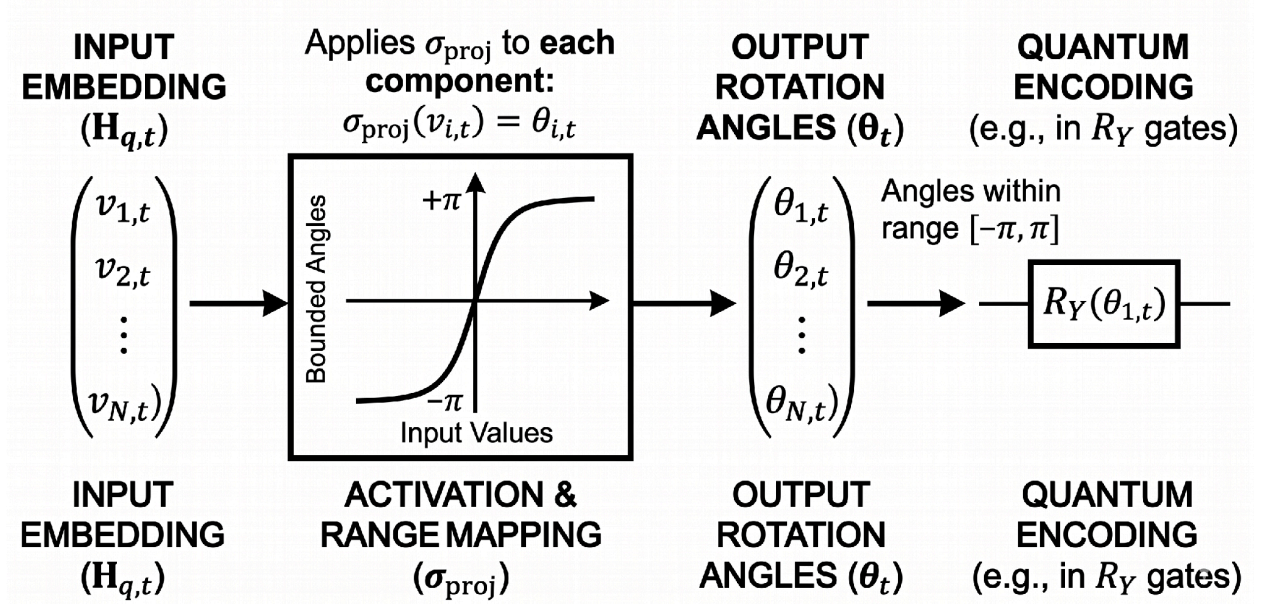
8. Pre-Projection

Embeddings are mapped to bounded angles suitable for encoding into quantum qubit rotations.

$$\alpha_t = \tanh(W_{in}[h_t \parallel w_{i,t} \parallel \tau_t] + b_{in})$$

Terms:

- h_t : Input embedding at time t .
- $\tanh$: Activation function mapping to range $[-1, 1]$.
- α_t : Vector of rotation angles for time t .



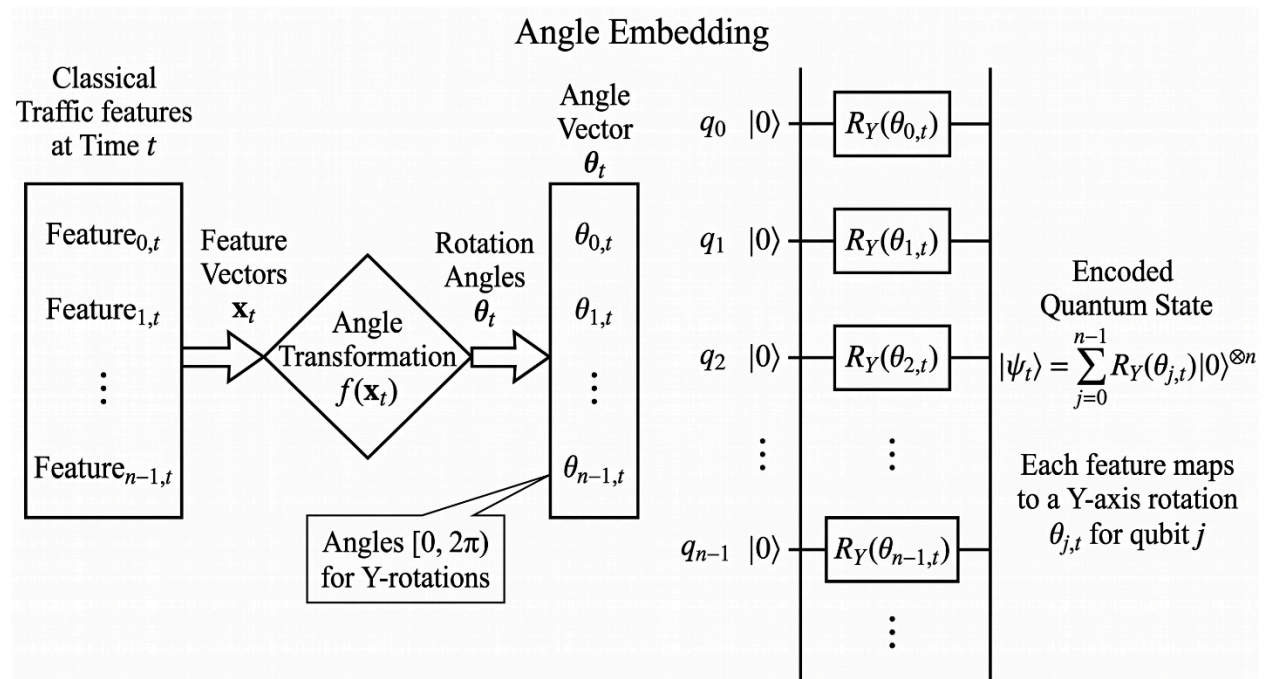
9. Angle Embedding

Classical data is encoded into quantum states via Y-axis rotations on qubits.

$$R_Y(\alpha_{q,t})|0\rangle$$

Explanation: Encodes the contextual traffic state into the quantum Hilbert space as a rotation

R_Y for qubit q at time t .



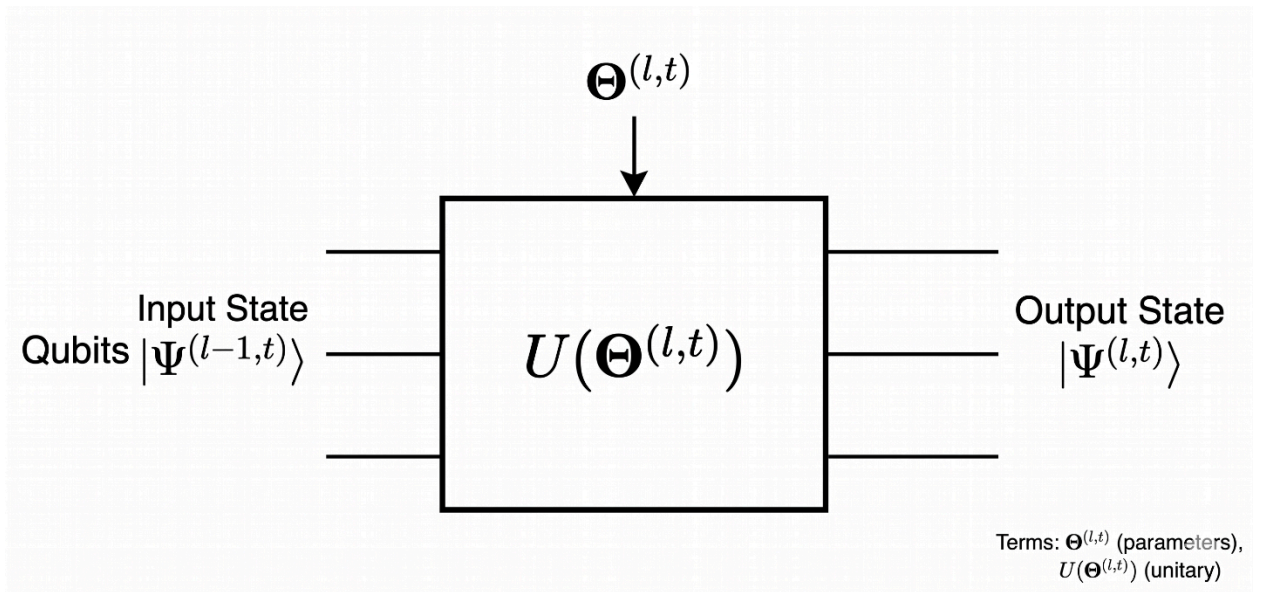
10. Variational Evolution

Trainable quantum gates explore high-dimensional correlations between traffic and environmental features.

$$U_{l,t} = e^{-i\theta_{l,2,t}Z} e^{-i\theta_{l,1,t}Y} e^{-i\theta_{l,0,t}Z}$$

Terms:

- $\theta_{l,k,t}$: Trainable parameters for layer l at time t .
- $U_{l,t}$: Unitary evolution operation.

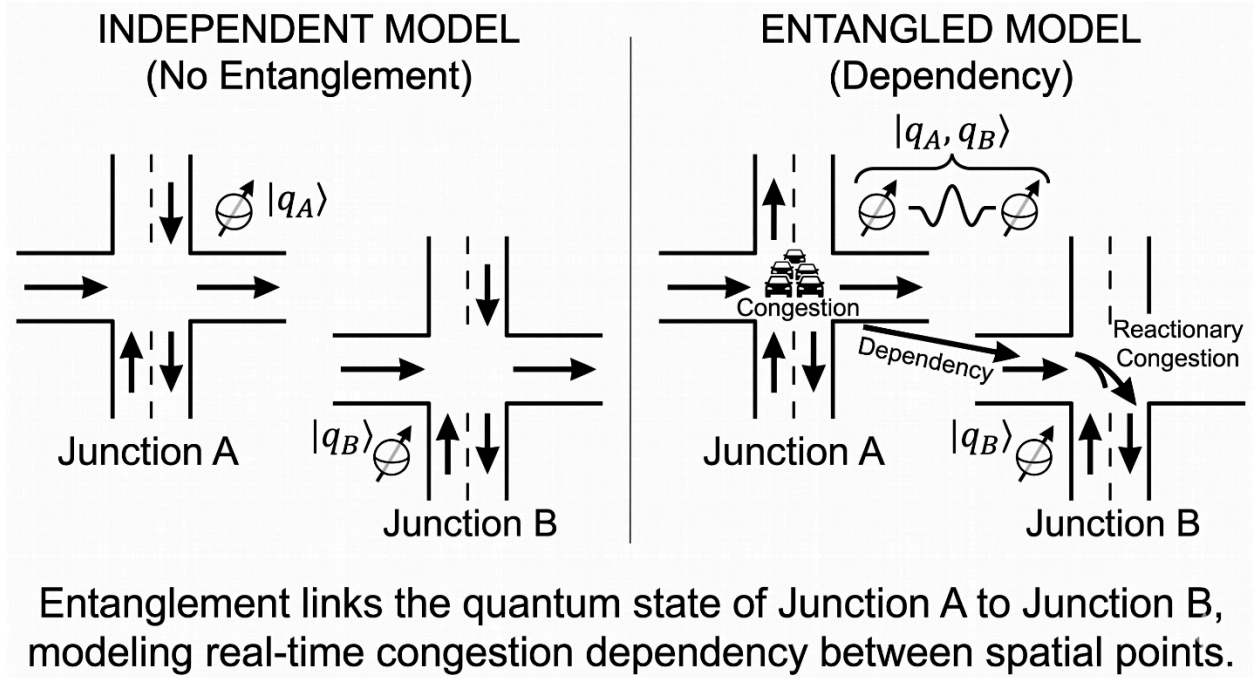


11. Entanglement

Links qubits to model the inter-dependencies between different spatial points in the city.

$$CNOT(Q_{0,t}, Q_{1,t}), CNOT(Q_{1,t}, Q_{2,t}), \dots$$

Explanation: Ensures the state of one qubit (representing a junction) is dependent on another, modeling complex congestion propagation.



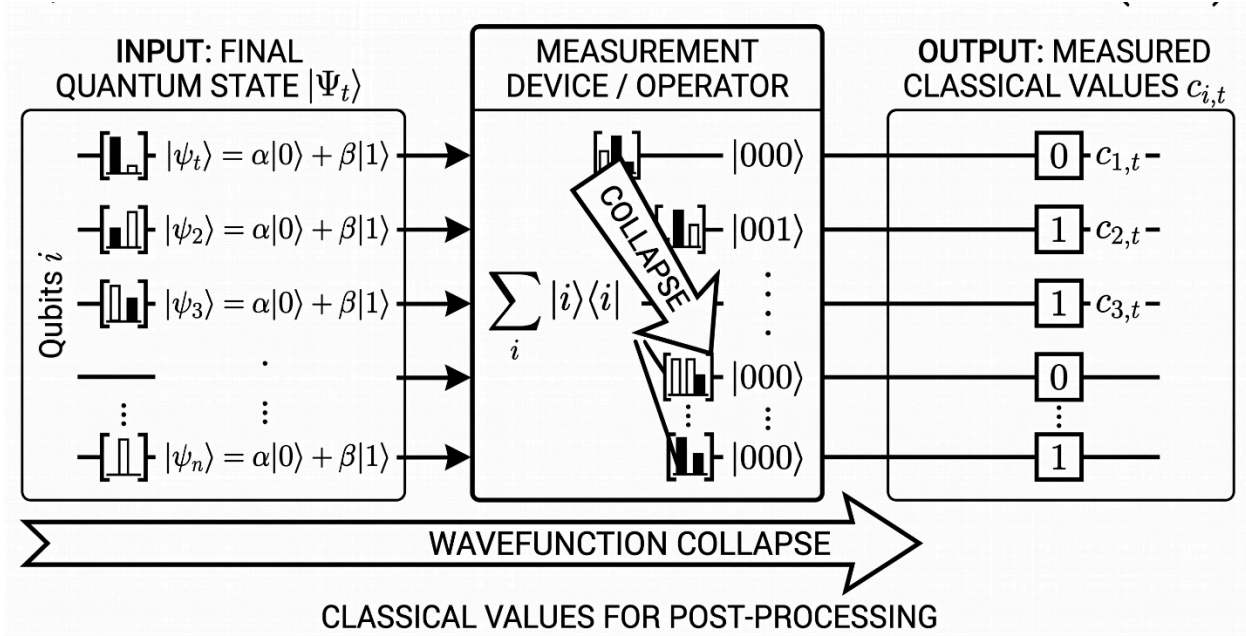
12. Measurement

The quantum state is collapsed into classical expectation values for post-processing.

$$o_{q,t} = \langle \psi_t | Z_q | \psi_t \rangle$$

Terms:

- $|\psi_t\rangle$: Final quantum state at time t .
- $o_{q,t}$: Measured classical value for qubit q .

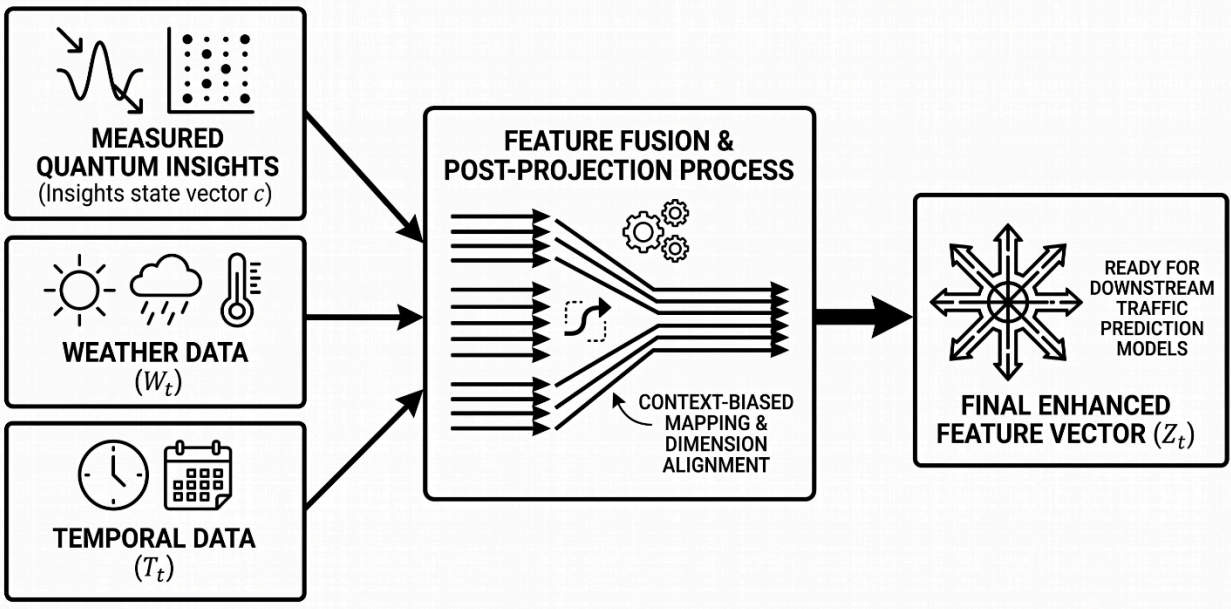


13. Post-Projection

The measured quantum insights are mapped back to the classical feature space, biased by current context.

$$q_t = W_{out}[o_t \parallel w_{avg,t} \parallel \tau_t] + b_{out}$$

Explanation: Fuses the quantum output with weather and temporal data to generate the final enhanced feature vector q_t .

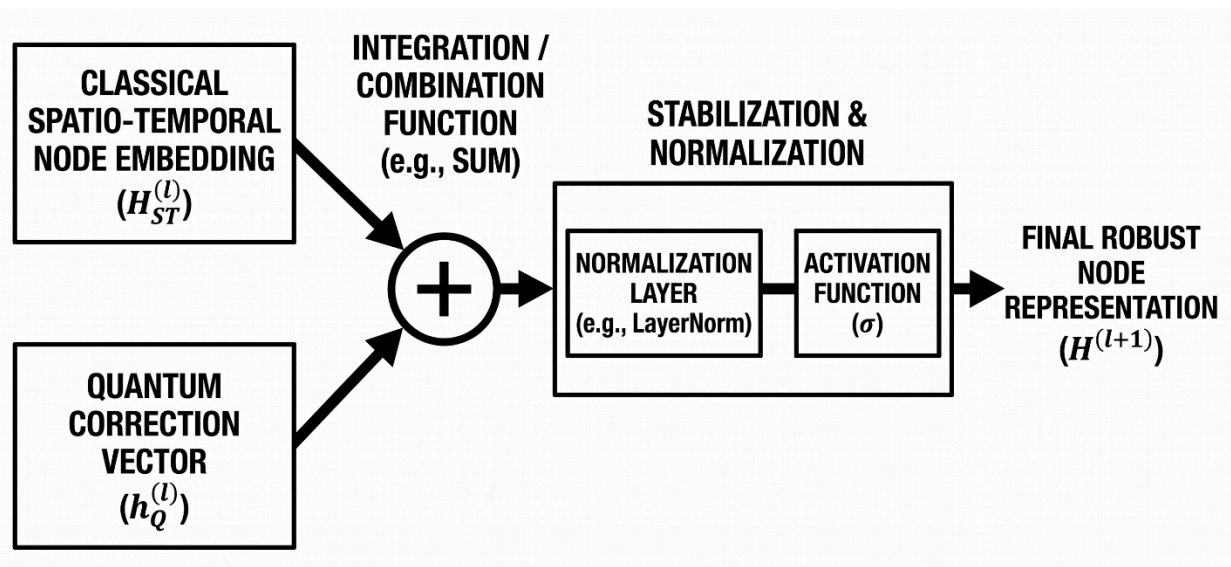


14. Quantum-Enhanced Update

The final node representation is stabilized and normalized to produce the most robust feature state.

$$h_{i,t}^{(1)} = \text{LayerNorm}(h_{i,t}^{(1)} + Q_{\Theta,t}(h_{i,t}^{(1)}))$$

Explanation: Integrates the quantum "correction" into the classical spatio-temporal embedding.



15. Edge Prediction (Future Forecast for $t + 1$)

The model predicts the traffic state for the next instance based on the current context and upcoming forecasts.

$$\hat{y}_{ij,t+1} = \sigma(MLP(h_{i,t} \parallel h_{j,t} \parallel f_{ij,t} \parallel \tau_t \parallel \tau_{t+1} \parallel w_{i,t} \parallel w_{i,t+1}))$$

Terms:

- $\hat{y}_{ij,t+1}$: Predicted traffic property for the road between nodes i, j at time $t + 1$.
- $\tau_{t+1}, w_{i,t+1}$: Forecasted time and weather parameters for the next instance.

EDGE TRAFFIC PREDICTION DIAGRAM (FROM t to $t + 1$)

